INDUCTIVE BIAS IN MACHINE LEARNING

How Learning Algorithms Generalize from Limited Data

Machine Learning — Short Answer Test

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Register No.: [REGISTER NUMBER]



LIMITED DATA → ASSUMPTIONS → GENERALIZATION

“Learning is not only about data — it is also about assumptions.”

WHY CAN'T A MODEL LEARN EVERYTHING FROM DATA?

A small training sample leaves many possible explanations.

OBSERVED EXAMPLES

Study Hours	Result
2 hrs	Fail
4 hrs	Pass
6 hrs	Pass

SAME DATA, DIFFERENT RULES

H_1 Passing requires ≥ 3 hours

H_2 Passing requires ≥ 4 hours

H_3 Passing depends on a more complex condition

UNSEEN CASE: 3.5 HOURS — **H_1: Pass** — **H_2: Fail**

Multiple hypotheses explain the same data, but disagree on unseen cases.

The data alone cannot resolve this.
Which hypothesis should the machine choose?

HOW DOES A MACHINE CHOOSE WHAT TO BELIEVE?

INDUCTIVE BIAS

The assumptions a learning algorithm uses to generalize from observed examples to unseen cases.

Hypothesis space = the set of candidate rules a learner can represent.

01 RESTRICTION BIAS

Limits which hypotheses are allowed

Example: allow only linear decision boundaries, so curved boundaries cannot be considered.

This reduces the number of candidate rules the learner must consider.

02 PREFERENCE BIAS

Ranks the hypotheses that are allowed

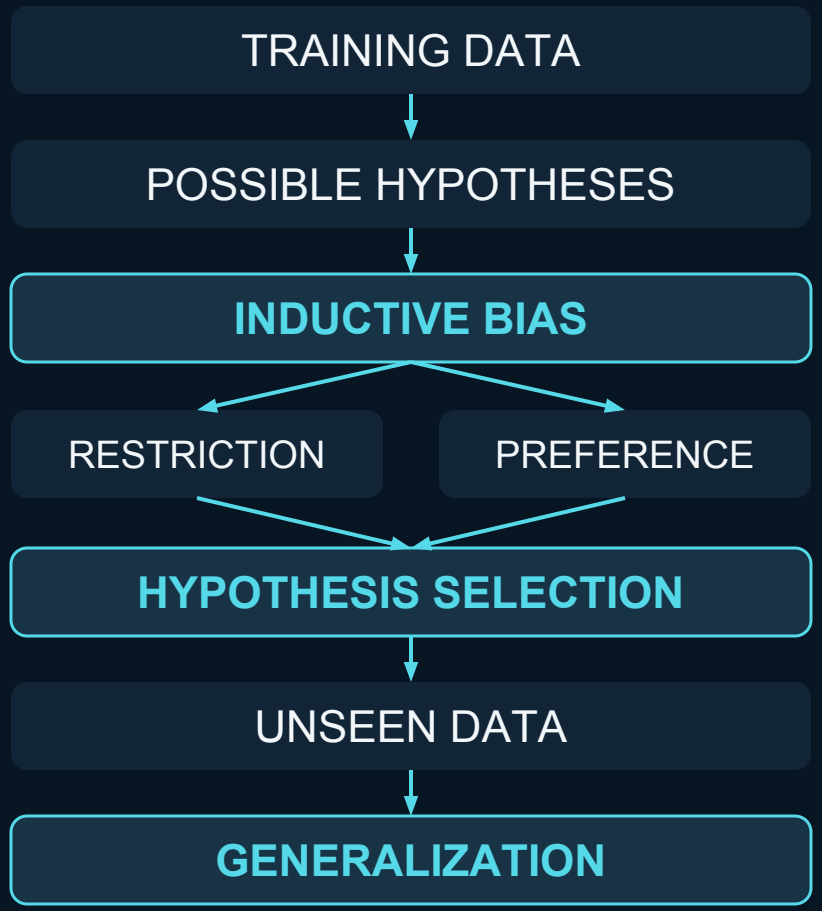
Example: prefer a simpler rule when several hypotheses explain the training examples.

This guides selection without excluding all other valid hypotheses.

Restriction = what the learner can consider

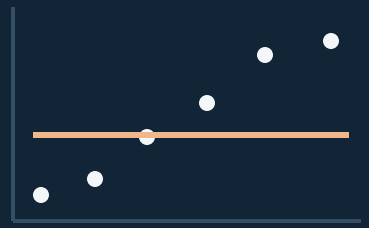
Preference = what the learner prefers

FROM TRAINING DATA TO UNSEEN DECISIONS



ASSUMPTIONS SHAPE MODEL BEHAVIOR

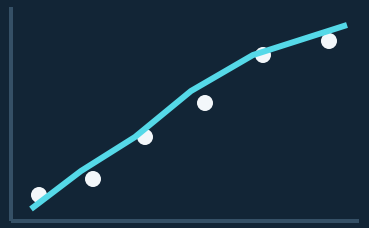
TOO RESTRICTIVE



Underfitting

Too simple to capture important patterns.

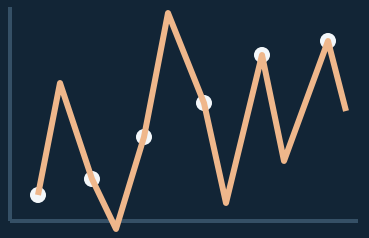
APPROPRIATE



Good generalization

Captures patterns that transfer to new data.

TOO FLEXIBLE



Overfitting

Fits noise and may fail on unseen examples.

Generalization means making accurate predictions on unseen data.

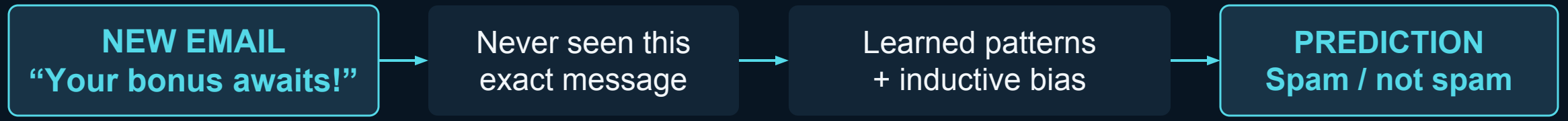
Bias helps when its assumptions fit the task.

FROM LIMITED DATA TO UNSEEN DECISIONS

EMAIL SPAM DETECTION

“Win a free prize!”	“Claim your reward now!”	“Meeting at 10 AM”
SPAM	**SPAM**	**NOT SPAM**

The model assumes that messages with similar relevant features tend to share labels.



FINAL TAKEAWAYS

Limited data
To predict unseen cases, the learner needs assumptions.

Assumptions
These assumptions guide which hypothesis it selects.

Good inductive bias
Bias can help when its assumptions match the learning problem.

“Data tells the model what it has seen. Inductive bias helps it decide what to expect next.”

THANKYOU